

EE2120: Distributed Smart Temperature Monitoring and Sensor Fusion Network

Phase 1: Technical Literature Review & Sensor Technology Selection

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Updated Sensor Selection Report

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1 Introduction and Project Scope

The EE2120 mini project requires a distributed temperature-monitoring node that can acquire temperature through two independent sensing channels and later combine those measurements using a sensor-fusion algorithm. The sensing subsystem must therefore provide measurement diversity rather than simple duplication.

The preliminary design selected one analog and one digital temperature-sensing architecture. Following procurement, hardware integration, and laboratory characterization, the final sensing pair used in the prototype is:

- **LM35DZ waterproof probe assembly** as the analog temperature channel.
- **DS18B20 waterproof digital temperature probe** as the digital temperature channel.

Both devices measure the same physical variable, but their signal paths are fundamentally different. The LM35DZ channel produces an analog voltage that must be digitized by the ESP32 ADC, while the DS18B20 performs temperature conversion internally and communicates through a digital 1-Wire interface. This difference creates the measurement diversity needed for later filtering, diagnostics, and sensor fusion.

The purpose of this report is to review alternative sensing technologies, document the final sensor selection, summarize the principal manufacturer specifications, and explain why the final LM35DZ–DS18B20 pairing is appropriate for the EE2120 system.

2 Comparative Evaluation of Candidate Temperature Sensors

Several temperature-sensing technologies were considered before finalizing the hardware configuration.

2.1 DHT22 Digital Temperature and Humidity Sensor

The DHT22 combines temperature and humidity measurement in a digital package. Although convenient for environmental monitoring, its relatively slow update rate makes it less suitable

for this project, where the sensor-fusion stage benefits from more frequent temperature observations. Its integrated digital interface also provides less architectural contrast if paired with another digital temperature sensor.

2.2 NTC/PTC Thermistors

Thermistors are inexpensive, compact, and can respond rapidly to changes in temperature. However, their resistance-temperature relationship is strongly nonlinear. Accurate conversion therefore requires either lookup tables or nonlinear equations such as the Steinhart–Hart model, together with a resistor network and analog-to-digital conversion. This increases calibration and firmware complexity for a student prototype.

2.3 PT100/PT1000 Resistance Temperature Detectors

Platinum RTDs provide excellent stability and linearity and are widely used in industrial instrumentation. However, they are resistive devices with relatively small signal changes and normally require a precision excitation source and signal-conditioning circuitry. Practical ESP32 interfacing therefore needs additional hardware such as an instrumentation amplifier or a dedicated RTD interface module, increasing cost and circuit complexity.

2.4 LM35 Family Analog Temperature Sensors

The LM35 family produces an output voltage directly proportional to temperature in degrees Celsius. The nominal transfer characteristic is approximately $10\text{ mV }^{\circ}\text{C}^{-1}$. This makes the sensor straightforward to interface with an ADC and avoids the strong nonlinearity associated with thermistors.

For the final prototype, the analog channel is implemented using an **LM35DZ waterproof probe assembly**. The designation LM35DZ refers to the LM35D-grade semiconductor in the TO-92 package; the waterproof construction is part of the purchased probe assembly rather than a separate manufacturer-defined LM35DZ electrical grade. The waterproof mechanical assembly is important because the completed system is intended for practical temperature measurements where direct contact with water or enclosed thermal environments may occur.

2.5 DS18B20 Digital Temperature Sensor

The DS18B20 integrates the temperature-sensing element, analog-to-digital conversion, digital data formatting, and 1-Wire communications within the sensor. It supports programmable 9-bit to 12-bit resolution and allows multiple uniquely addressed devices to share one 1-Wire bus.

For the final prototype, a **waterproof DS18B20 probe** is used. The electrical limits quoted in this report are those of the DS18B20 device; the sealed metal probe construction is the installed assembly. The sealed probe construction provides mechanical protection and simplifies immersion testing.

3 Detailed Review of the Selected Sensors

3.1 LM35DZ Waterproof Probe Assembly

The LM35DZ is an analog integrated-circuit temperature sensor whose output voltage is proportional to Celsius temperature. The underlying LM35 family is specified with a nominal scale factor of $10 \text{ mV } ^\circ\text{C}^{-1}$. For the LM35D/LM35DZ device, the manufacturer specifies operation from a 4 V to 30 V supply, with the LM35D grade rated over approximately 0°C to 100°C . At 25°C , the LM35D typical accuracy is approximately $\pm 0.6^\circ\text{C}$.

Signal path:

$$T \rightarrow V_{\text{LM35DZ}} \rightarrow \text{ESP32 ADC} \rightarrow \text{digital temperature estimate}$$

Because the output is analog, the complete measurement accuracy depends not only on the sensor but also on the ESP32 ADC, supply quality, wiring, grounding, electrical noise, and the calibration of the installed channel.

The final hardware therefore treats the LM35DZ probe as a complete *installed measurement channel*, rather than assuming that the nominal $10 \text{ mV } ^\circ\text{C}^{-1}$ transfer function alone is sufficient.

Laboratory characterization of the assembled channel produced the empirical calibration relation

$$T_{\text{LM35DZ}} = 0.081567 (\text{ADC}) + 14.5187 \quad (1)$$

over the tested range from approximately 29.375°C to 60.5625°C .

The fitted relationship gave

$$R^2 = 0.999828$$

with a mean absolute calibration error of approximately

$$\text{MAE} = 0.0810^\circ\text{C}.$$

These results confirm that the installed analog channel is highly linear after calibration, although it remains affected by ADC noise.

3.2 DS18B20 Waterproof Digital Probe

The DS18B20 is a digital thermometer using the 1-Wire protocol. It supports 9-bit, 10-bit, 11-bit, and 12-bit conversion modes. The corresponding nominal temperature increments are:

Resolution	Temperature Increment
9-bit	0.5°C
10-bit	0.25°C
11-bit	0.125°C
12-bit	0.0625°C

At 12-bit resolution, the maximum conversion time is approximately 750 ms. The device communicates through a 1-Wire open-drain data line and therefore requires an external pull-up resistor. The datasheet specifies an approximately $5 \text{ k}\Omega$ pull-up for the 1-Wire bus and also illustrates $4.7 \text{ k}\Omega$ implementations. In the final ESP32 circuit, the DS18B20 data line is connected to GPIO4 with a $4.7 \text{ k}\Omega$ pull-up to 3.3 V.

The digital architecture reduces sensitivity to analog wiring noise and avoids use of the ESP32 ADC. However, resolution settings, conversion time, probe construction, and thermal encapsulation still affect the behavior of the complete installed channel.

4 Manufacturer-Level Technical Specification Comparison

Parameter	LM35DZ Waterproof Analog Channel	DS18B20 Waterproof Digital Channel
Measurement principle	Analog voltage proportional to temperature	Integrated digital thermometer
Communication / output	Continuous analog output, nominally $10 \text{ mV } ^\circ\text{C}^{-1}$	Digital 1-Wire interface
Typical supply used in prototype	5 V	3.3 V
Manufacturer supply capability	LM35D/LM35DZ: 4 V to 30 V	3.0 V to 5.5 V
Rated temperature range	LM35D/LM35DZ: 0°C to 100°C	-55°C to 125°C
Nominal accuracy	LM35D/LM35DZ: approximately $\pm 0.6^\circ\text{C}$ typical at 25°C	$\pm 0.5^\circ\text{C}$ from -10°C to 85°C
Resolution	Determined by sensor slope, ESP32 ADC, noise, and calibration	Programmable 9-bit to 12-bit; 0.5°C to 0.0625°C per code
Digital conversion delay	Not applicable inside the LM35DZ; ADC conversion is performed by ESP32	Up to approximately 750 ms at 12-bit resolution
External interface components	ADC input, appropriate supply decoupling/filtering, common ground	1-Wire bus with external pull-up (4.7 k Ω used in prototype)

Table 1: Technical comparison of the final sensor pair.

5 Measured Characterization of the Final Selected Sensors

The initial literature-based design rationale assumed that the analog temperature sensor would provide a faster transient response than the digital channel because the LM35 does not contain the DS18B20’s internal digital conversion process. Laboratory testing of the *actual waterproof probe assemblies* showed that this assumption does not hold for the completed hardware.

5.1 Static Calibration

Using the DS18B20 as the reference channel during the gradual temperature sweep, the LM35DZ channel produced the calibrated model

$$T_{\text{LM35DZ}} = 0.081567 (\text{ADC}) + 14.5187.$$

The coefficient of determination was

$$R^2 = 0.999828,$$

and the mean absolute tracking error relative to the DS18B20 reference was approximately

$$0.0810\text{ }^{\circ}\text{C}.$$

This indicates very strong static agreement after channel calibration.

5.2 Steady-State Noise

During the steady-state noise experiment, the calibrated LM35DZ channel had a sample standard deviation of approximately

$$\sigma_{\text{LM35DZ}} = 0.5425\text{ }^{\circ}\text{C}.$$

The DS18B20 readings remained at a single $0.5\text{ }^{\circ}\text{C}$ code during the 9-bit test, giving an observed sample standard deviation of

$$\sigma_{\text{DS18B20}} = 0.0000\text{ }^{\circ}\text{C}.$$

This zero observed variance must not be interpreted as zero physical measurement uncertainty. It is primarily a consequence of quantization during a thermally stable interval.

5.3 Thermal Step Response

The final hot-water step test measured the 63.2% response time constants of the waterproof probe assemblies as approximately

$$\tau_{\text{LM35DZ}} = 13.96\text{ s}$$

and

$$\tau_{\text{DS18B20}} = 8.04\text{ s}.$$

Therefore, the installed DS18B20 waterproof probe responded faster than the installed LM35DZ waterproof probe assembly in the experiment.

This result is significant. It demonstrates that the overall thermal response of the final sensor channel is governed not only by the internal sensing technology but also by the mechanical probe construction, encapsulation, thermal mass, mounting, and heat-transfer path.

Consequently, sensor-fusion logic must be based on the experimentally characterized *complete probes*, not on generic assumptions about bare sensor dies.

6 Engineering Rationale for the Final Sensor Pair

The final LM35DZ waterproof probe and DS18B20 waterproof probe provide a useful heterogeneous pair for the EE2120 project.

6.1 Measurement Diversity

The two channels use different electrical measurement paths:

- The LM35DZ channel is analog and depends on the ESP32 ADC.
- The DS18B20 channel performs conversion internally and communicates digitally.

This diversity enables independent fault detection and comparison between analog and digital signal chains.

6.2 Complementary Noise Characteristics

The experimental results show that the LM35DZ channel exhibits substantially greater short-term variation because its measurement passes through the analog signal path and ESP32 ADC.

The DS18B20 channel is much more stable during steady-state operation, although its quantized output should still be assigned a non-zero uncertainty in any fusion algorithm.

6.3 Measured Dynamic Behavior

The earlier literature-only assumption that the analog channel would necessarily be the faster channel has been rejected by the experimental results.

For the installed waterproof probes:

$$\tau_{DS18B20} < \tau_{LM35DZ}.$$

Therefore, during rapid thermal transitions, the final sensor-fusion algorithm should use the measured probe dynamics rather than automatically assigning greater transient weight to the analog channel.

6.4 Suitability for Sensor Fusion

The pairing remains highly suitable for sensor fusion because the two sensors provide:

- independent measurement architectures,
- different noise behavior,
- different quantization characteristics,
- different transient responses,
- separate failure modes,
- and independent raw measurements for diagnostics.

A practical fusion algorithm can therefore combine steady-state stability, calibrated analog information, dynamic response modeling, and measurement covariance estimates.

7 Final Sensor Selection

Based on the literature review, hardware availability, electrical compatibility, waterproof construction, ESP32 interfacing requirements, and subsequent experimental characterization, the final temperature-sensor pair for the EE2120 node is:

Channel	Selected Device	Interface
Analog	LM35DZ waterproof probe assembly	ESP32 ADC1, GPIO34
Digital	DS18B20 waterproof temperature probe	1-Wire, GPIO4

The LM35DZ channel requires empirical calibration before temperature values are used in later processing. The verified calibration equation for the characterized prototype is

$$T_{\text{LM35DZ}} = 0.081567 (\text{ADC}) + 14.5187$$

within the experimentally tested range.

The DS18B20 provides the more stable steady-state measurement in the recorded tests and also exhibited the faster probe-level transient response in the hot-water step experiment.

8 Conclusion

The updated literature review confirms the use of the **LM35DZ waterproof analog probe assembly** and the **DS18B20 waterproof digital probe** as the final sensing pair for the EE2120 distributed temperature-monitoring node.

The original motivation for choosing heterogeneous analog and digital sensors remains valid: the two channels provide independent signal paths and complementary measurement behavior suitable for sensor fusion.

However, characterization of the completed waterproof probe assemblies changed one important design conclusion. The final hardware does *not* support the assumption that the analog probe is dynamically faster. The measured time constants were approximately 13.96 s for the LM35DZ assembly and 8.04 s for the DS18B20 probe. The final sensor-fusion design must therefore use measured channel characteristics rather than generic component-level expectations.

The LM35DZ channel is highly linear after empirical calibration but requires filtering because of analog and ADC noise. The DS18B20 channel provides stable digital measurements, but its quantization and finite accuracy require a non-zero measurement uncertainty in fusion calculations.

The selected pair therefore provides a suitable basis for the remaining EE2120 tasks: calibrated data acquisition, adaptive or model-based sensor fusion, WiFi/GPS integration, SCADA visualization, and distributed temperature estimation.

References

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